THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapter 3: **The World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** terminating	denominator 8 is of the form 2^a5^b
2	**(a)** terminating	denominator 40 is of the form 2^a5^b
3	**(c)** $\frac{5}{11}$	standard method ⇒ $\frac{5}{11}$
4	**(b)** $\frac{161}{450}$	standard method ⇒ $\frac{161}{450}$
5	**(a)** irrational	non-terminating and non-repeating
6	**(b)** 3	number of places $= \max(a, b)$ in 2^a5^b
7	**(b)** 56	Brahmagupta's rules for signs
8	**(a)** −7	−7 is an integer
9	**(d)** Hippasus	as per the chapter
10	**(a)** False; $\sqrt{2} \times \sqrt{2} = 2$ is rational	product of irrationals can be rational

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let $x = 0.\overline{6}$. Then $10x = 6.\overline{6}$; subtracting, $9x = 6 \Rightarrow x = \frac{2}{3}$	2
12	Let $x = 0.\overline{45}$. Then $100x = 45.\overline{45}$; subtracting, $99x = 45 \Rightarrow x = \frac{5}{11}$	2
13	Let $x = 0.1\overline{3}$. $10x = 1.\overline{3}$ and $100x = 13.\overline{3}$; subtracting, $90x = 12 \Rightarrow x = \frac{2}{15}$	2
14	Let $x = 0.\overline{18}$. Then $100x = 18.\overline{18}$; subtracting, $99x = 18 \Rightarrow x = \frac{2}{11}$	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Assume $\sqrt{2} = \frac{p}{q}$ with p, q co-prime, $q \neq 0$ (1). $2q^2 = p^2 \Rightarrow p^2$ even $\Rightarrow p$ even; $p = 2k$ (1). Then $q^2 = 2k^2 \Rightarrow q$ even — so 2 divides both p and q, contradicting co-primality. Hence $\sqrt{2}$ is irrational (1).	1+1+1
16	Assume $\sqrt{3} = \frac{p}{q}$ (co-prime). $3q^2 = p^2 \Rightarrow 3 \mid p$; $p = 3k \Rightarrow q^2 = 3k^2 \Rightarrow 3 \mid q$. Both divisible by 3 — contradiction. So $\sqrt{3}$ is irrational.	1+1+1
17	Assume $\sqrt{5} = \frac{p}{q}$ in lowest terms. $5q^2 = p^2 \Rightarrow 5 \mid p \Rightarrow p = 5k \Rightarrow q^2 = 5k^2 \Rightarrow 5 \mid q$. Contradiction ⇒ $\sqrt{5}$ is irrational.	1+1+1
	OR: If $3 + \sqrt{2} = r$ (rational), then $\sqrt{2} = r - 3$ would be rational — contradiction. Hence $3 + \sqrt{2}$ is irrational.	1+1+1
18	Let $x = 2.3\overline{57}$. $10x = 23.\overline{57}$; $1000x = 2357.\overline{57}$. Subtract: $990x = 2334 \Rightarrow x = \frac{2334}{990} = \frac{389}{165}$.	1+1+1
	OR: $x = 0.16\overline{6}$; $10x = 1.6\overline{6}$; $100x = 16.6\overline{6}$. $90x = 15 \Rightarrow x = \frac{1}{6}$.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Contradiction proof: $\sqrt{2} = \frac{p}{q}$, $2q^2 = p^2$, p and q both even — contradiction (3). (ii) If $3\sqrt{2} - 5 = r$ (rational) then $\sqrt{2} = \frac{r+5}{3}$ rational — contradiction (2).	3+2
	OR: (i) $0.\overline{27} = \frac{27}{99} = \frac{3}{11}$. $x = 1.2\overline{5}$: $10x = 12.\overline{5}$, $100x = 125.\overline{5}$; $90x = 113 \Rightarrow x = \frac{113}{90}$. (ii) $9x = 9 \Rightarrow x = 1$: so $0.\overline{9} = 1$ exactly; decimal forms are not unique.	2+2+1
20	(i) All are primes between 10 and 20; next primes 23, 29, 31. (ii) $\frac{15}{2} \times 12 = 90$ ingots. (iii) No: $3 - 5 = -2$ is not a natural number.	2+2+1
	OR: (i) $80 = 2^4 \times 5 \Rightarrow \frac{13}{80}$ terminates; $12 = 2^2 \times 3$ and $343 = 7^3 \Rightarrow$ repeating. (ii) $\frac{13}{80} = \frac{13 \times 125}{10^4} = 0.1625$. (iii) $\frac{5}{12} = 0.41666\ldots = 0.41\overline{6}$; repeating digit 6.	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 3: **The World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** non-terminating repeating	denominator 6 has a prime factor other than 2 and 5
2	**(b)** non-terminating repeating	denominator 14 has a prime factor other than 2 and 5
3	**(b)** $\frac{1}{3}$	standard method ⇒ $\frac{1}{3}$
4	**(a)** rational	$= 9$
5	**(d)** irrational	not a ratio of integers
6	**(a)** 2	number of places $= \max(a, b)$ in $2^a 5^b$
7	**(a)** 14	Brahmagupta's rules for signs
8	**(a)** $\frac{5}{8}$	$\frac{1}{2}\left(\frac{1}{2} + \frac{3}{4}\right) = \frac{5}{8}$
9	**(d)** subtracting a number from itself ($a - a = 0$)	Brāhmasphuṭasiddhānta (628 CE)
10	**(b)** infinitely many rational numbers	density of $\mathbb{Q}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	8 has prime factors only 2 and/or 5 ($2^a 5^b$) ⇒ terminating	1+1
12	12 has a prime factor other than 2 and 5 ⇒ non-terminating repeating	1+1
13	20 has prime factors only 2 and/or 5 ($2^a 5^b$) ⇒ terminating	1+1
14	9 has a prime factor other than 2 and 5 ⇒ non-terminating repeating	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = 2.45\overline{37}$. $100x = 245.\overline{37}$; $10000x = 24537.\overline{37}$. $9900x = 24292 \Rightarrow x = \frac{24292}{9900} = \frac{6073}{2475}$.	1+1+1
16	$100x = 45.\overline{45}$, $99x = 45 \Rightarrow x = \frac{45}{99} = \frac{5}{11}$. Check: $5 \div 11 = 0.4545\ldots$ ✓	2+1
17	Common denominator 24: $\frac{8}{24}$ and $\frac{12}{24}$. Take $\frac{9}{24}, \frac{10}{24}, \frac{11}{24}$, i.e. $\frac{3}{8}, \frac{5}{12}, \frac{11}{24}$.	1+1+1
	OR: Write $3 = \frac{18}{6}$ and $4 = \frac{24}{6}$: $\frac{19}{6}, \frac{20}{6}, \frac{21}{6}, \frac{22}{6}, \frac{23}{6}$ (any five valid).	1+1+1
18	e.g. 3.14152, 3.14155, 3.14158.	1+1+1
	OR: $\frac{77}{210} = \frac{11}{30}$; $30 = 2 \times 3 \times 5$ has factor 3 ⇒ non-terminating repeating.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 7 kurtas; left 1 m. (ii) $12.5 \div 2 = 6.25$: 6 kurtas; left 0.5 m $= \frac{1}{2}$ m. (iii) Left: 1 m and $\frac{1}{2}$ m; total $\frac{3}{2}$ m.	2+2+1
	OR: (i) $\sqrt{3} = \frac{p}{q}$; $3q^2 = p^2 \Rightarrow 3 \mid p$, $p = 3k \Rightarrow 3 \mid q$; contradiction (3). (ii) If $2 - \sqrt{3} = r$, $\sqrt{3} = 2 - r$ rational — contradiction (2).	3+2
20	$\frac{3}{5} = \frac{18}{30}$, $\frac{4}{5} = \frac{24}{30} \Rightarrow \frac{19}{30}, \frac{20}{30}, \frac{21}{30}, \frac{22}{30}, \frac{23}{30}$ (3). To fit 5 integers strictly between k_1 and k_2 we need $k_2 - k_1 > 6$ — i.e. ≥ 7 (2). (With denominator 30 the difference is 6; the five values lie strictly between.)	3+2
	OR: (i) $\sqrt{2}$. (ii) Draw $OA = 1$, perpendicular $AB = 1$, then arc of radius OB meets the line at $P = \sqrt{2}$. (iii) Proof by contradiction shows $\sqrt{2}$ is irrational. (iv) $1.4142\ldots$ non-terminating, non-repeating.	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.